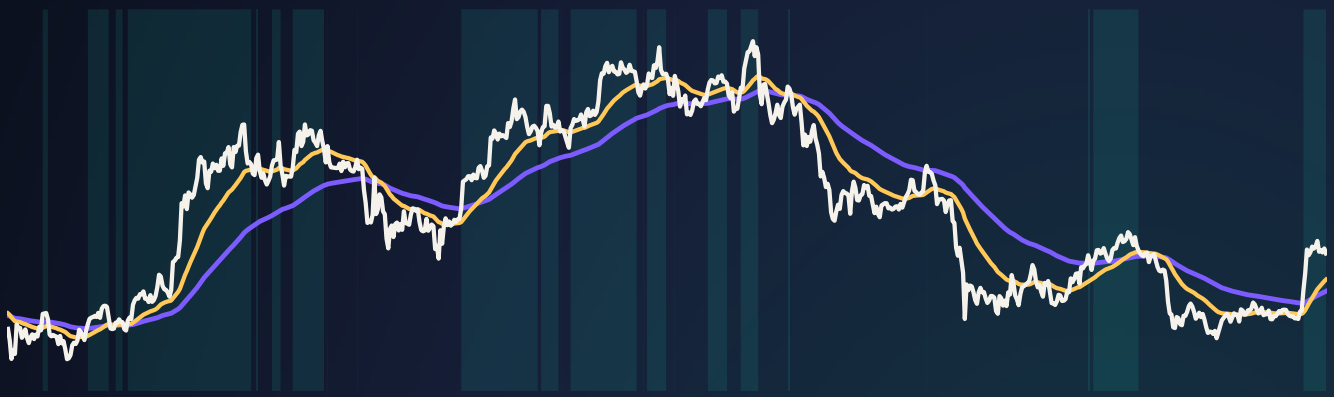
RULE TEST #01

How this strategy was tested.

The full record: every test it had to survive, every number it produced, and the market it failed on. Written for someone with no quant background - every technical term is translated on the page where it first appears.



BTC daily closes. Teal = holding, dark = in cash.

+54.2% CAGR	**-46.7%** MAX DRAWDOWN	**34x** TOTAL RETURN
0.0055 P-VALUE	**83** TRADES	**30.1%** WIN RATE

Personal experience. Not financial advice. Nothing here is a recommendation to trade.

START HERE

The road map.

Six stages, in the order they happened. Nothing below was decided after seeing the results it produced.

01 **The rule**
What it actually does, in one sentence and then in full.

02 **The trap**
140 versions were tested. Why that is a problem, not a boast.

03 **The setup**
What was locked down before any searching began.

04 **Five ways to break it**
Reality Check, unseen era, other coins, random timing, reshuffles.

05 **The results**
Every number, including the one that failed.

06 **The limits**
What this still does not prove.

How to read this

Every technical term is explained in a teal-edged box the first time it is used, and all of them are collected in the glossary on the next two pages. You can skip the glossary and come back to it.

GLOSSARY

Every term, in plain English.

You do not need any of this to follow the report - it is repeated in context wherever it matters.

Backtest

Running a trading rule over old price data to see what it would have done. It is a simulation, not a track record - nobody actually risked money.

Moving average

The average price over the last N days, recalculated daily. It smooths out the daily noise so you can see the direction underneath. A 100-day average moves slowly; a 34-day one reacts faster.

EMA

Exponential moving average. Same idea, but recent days count for more than old ones, so it turns sooner.

Regime filter

The slow line, used as a yes/no gate: is this market broadly trending up? If no, don't trade at all.

Trailing exit

The fast line, used as a moving stop. It follows price up, and you leave when price falls back through it.

CAGR

Compound annual growth rate. The single yearly return that, repeated, gets you from the start to the end. A way to compare periods of different lengths fairly.

Drawdown

How far you fell from your highest point before recovering. If $100k drops to $53k, that is a 47% drawdown. This is the number you actually have to live through.

The measurement vocabulary.

Buy & hold

Just buying and doing nothing. The benchmark every strategy has to beat, because doing nothing is free.

Win rate

The share of trades that made money. A low win rate is not automatically bad - this rule wins 3 times in 10 and still makes money, because the winners are much larger than the losers.

Sharpe ratio

Return divided by how bumpy the ride was. Higher means you got the same return with less violence. Roughly: below 0.5 is poor, around 1.0 is solid, above 2.0 is rare and usually too good to be true.

In-sample

The data you used to build and choose your rule. Results here are flattering by construction.

Out-of-sample

Data you deliberately never looked at while building, kept back to test the finished rule once. The only results that mean much.

Overfitting

Building a rule that describes the past perfectly and predicts nothing. Usually caused by trying many variations and keeping whichever fit best.

GLOSSARY, CONTINUED

The testing vocabulary.

Data mining

Testing lots of ideas and reporting the winner as if it were the only one you tried. The single most common way backtests lie.

p-value

The chance of getting a result this good if the idea were actually worthless. Small is good. Below 0.05 (5%) is the usual bar; this rule came in at 0.0055, about half a percent.

Monte Carlo

Re-running the same thing thousands of times with one element randomised, to see the range of outcomes rather than the single one you happened to get.

White's Reality Check

A test built specifically for the data-mining problem. It asks how good the luckiest of 140 worthless strategies would have looked, and makes your result beat that bar rather than beat zero.

Walk-forward

Re-choosing the settings each year using only what was known at the time, then trading the next year on that choice. It scores the method, not one lucky setting.

Round trip

The total cost of one complete trade - the fee to get in plus the fee to get out, quoted as one number because a trade always has both ends. 0.075% round trip means a $10,000 position costs about $7.50 in fees for the whole trade. It matters enormously for a rule that trades daily, and barely at all for one that trades ten times a year.

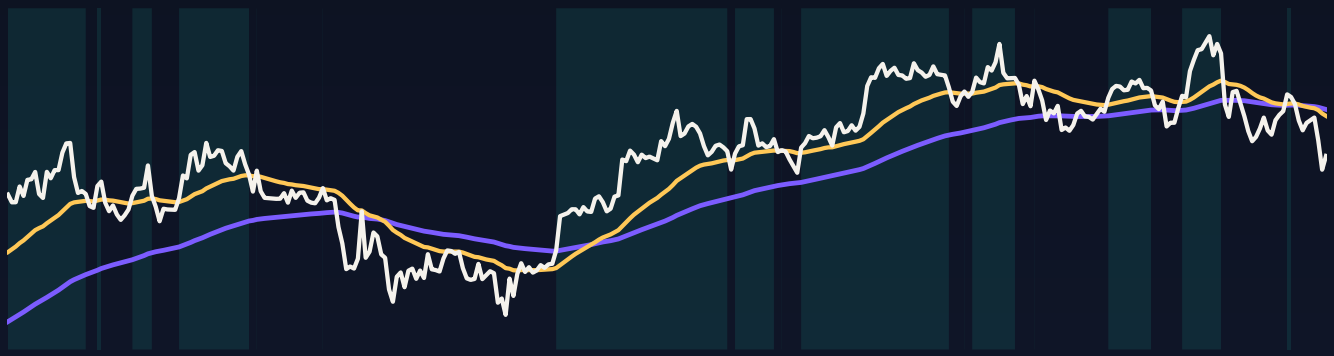
STAGE 01

The rule.

One sentence. Everything after this page is that sentence unpacked.

At each daily close, if price is above BOTH the 100-day and the 34-day average, be long at the next open.

If it is below EITHER one, be flat at the next open.



White = BTC price. Purple = 100-day average. Amber = 34-day average. Teal = holding.

MARKET	Bitcoin (BTC/USDT), spot or unleveraged perpetual
CANDLE	Daily, 00:00-23:59 IST
INDICATORS	100-day EMA = regime filter. 34-day EMA = trailing exit
SIGNAL CHECK	Once per day, at the daily close
EXECUTION	The next day's open - never the close you just read
ENTER LONG	close > 100-day AND close > 34-day
EXIT TO FLAT	close < 34-day OR close < 100-day
BETWEEN THEM	Flat. Wait for a close back above both
SHORTS	Never
SIZE / LEVERAGE	Full balance, 1x, no leverage
STOP LOSS	None. The 34-day average is the stop
TAKE PROFIT	None
COSTS ASSUMED	0.075% round trip

No indicators beyond two averages. No news. No discretion. No predictions.

STAGE 01

Two states, and nothing else.

There is no third thing to decide. You are either long or in cash.

IN PLAIN ENGLISH · REGIME FILTER AND TRAILING EXIT

The slow 100-day line answers 'is this market broadly going up?' The fast 34-day line answers 'is that move still alive?' One gates you in, the other takes you out.

LONG

about 43% of days

- Entered because close > 100-day AND > 34-day
- Hold while both stay true - no daily action needed
- Leave the moment either one fails

FLAT

about 57% of days

- Price is below the 34-day, the 100-day, or both
- Hold cash. No short, no hedge, no smaller position
- Re-enter only on a close back above both

Being flat over half the time is the point, not a flaw.

It is where the entire drawdown reduction comes from. In 2022 the rule was in the market on 3% of days - which is why 2022 cost it 6% while Bitcoin itself fell 64%.



A representative year: about 10 trades, four months holding, eight in cash.

STAGE 02

140 versions. That is the problem.

The best of 140 tries is not evidence. It is the expected outcome of trying 140 things, whether or not any of them work.

IN PLAIN ENGLISH · DATA MINING

If you test enough variations and report only the winner, you will always have a winner - even when every variation is worthless. The result measures how many tries you had, not whether the idea works.

What the 140 actually were

The search was a grid. Four settings, every combination tested:

- **5** fast averages
  8, 13, 21, 34, 50 days
- **7** slow averages
  100, 150, 175, 200, 225, 250, 300 days
- **2** exit modes
  fast line only, or either line
- **2** direction modes
  long only, or long and short

5 x 7 x 2 x 2 = 140 candidates



Every single one of the 140 made money.

That is the tell. A grid where every cell is profitable is not 140 pieces of evidence - it is one idea, trend-following, measured 140 different ways. The right conclusion is that the mechanism is robust and the exact settings are noise; the wrong one is that the best cell is special.

Per-candidate results are not reproduced: the original scripts and data no longer exist.

STAGE 03

Locked before any searching.

Choices made after seeing results are not findings. These were fixed first, and written down before the search ran.

IN PLAIN ENGLISH · IN-SAMPLE VS OUT-OF-SAMPLE

In-sample data is what you used to build the rule - results there flatter you by construction. Out-of-sample data is held back, unseen, and run once at the end. Look at it twice and you start tuning to it, and it stops being a test.

1. The session was fixed, not chosen

A daily rule needs a definition of 'a day'. Pick the timezone after seeing results and you have quietly tuned a parameter. So all 48 half-hour day-start offsets were tested first: all 48 were profitable and all 48 beat buy-and-hold. IST ranks 25th of 48, dead middle. It was picked because it is the close he can actually watch.



2. The data was split and sealed

SELECTION BTC 2018-2026	Used to search all 140 and pick a winner	
VALIDATION A BTC 2013-2017	Bitfinex. Untouched until the rule was final	SEALED
VALIDATION B 5 other assets	ETH, BNB, DOGE, SOL, XRP. One shot each	SEALED

3. The scoring metric was fixed

Sharpe ratio, declared before the search. Not total return - otherwise the winner is always whichever version took the most risk.

STAGE 04 · TEST 1 OF 5

White's Reality Check.

The test built specifically for the 140-versions problem.

IN PLAIN ENGLISH · WHITE'S REALITY CHECK

Ordinary statistics ask 'could this happen by chance?' That is the wrong question after 140 tries. This asks a harder one: if all 140 versions were worthless, how good would the luckiest of them have looked? Your winner has to beat THAT bar, not zero.

How it works, step by step

1. Take the real best-of-140 result.
2. Scramble the data so any real signal is destroyed but its statistical shape survives.
3. Run the identical 140-candidate search on the scrambled data. Record the best.
4. Repeat hundreds of times. You now have a distribution of 'best of 140, by luck alone'.
5. Ask what share of those lucky bests beat the real one.



p = 0.0055

Fewer than 6 lucky searches in 1,000 would have produced a winner this good. The usual bar is 50 in 1,000.

What it does not say

This clears the search procedure of having found pure noise. It does not prove the rule will keep working. That is what the next four tests are for.

STAGE 04 · TEST 2 OF 5

An era the search had never seen.

Bitcoin 2013-2017, from Bitfinex. Binance did not exist yet, so this data could not have leaked into the search even by accident.

IN PLAIN ENGLISH · OUT-OF-SAMPLE TEST

The rule was already final when this ran, and no settings were changed afterwards. That is the entire source of the test's power - and it is spent after one look.



Three regimes the search never saw: the 2013 bubble from $125 to $1,150, the 2014-15 collapse of 85%, and the 2016-17 rally. Log scale, real Bitfinex daily closes.

36	+72.9%	-39.6%
TRADES	CAGR	MAX DD
1.46	**$71,535**	**$58,596**
SHARPE	FROM $10k	BUY & HOLD

It beat buy-and-hold, through the worst bear market in the asset's history.

The honest caveat: a Sharpe of 1.46 here is HIGHER than the 1.31 the rule scored on the data it was built on. That is unusual, and it is luck rather than skill - 2013-2017 was an exceptionally trendy era. The number to anchor on is the 1.00 average across all validation sets, not this one.

STAGE 04 · TEST 3 OF 5

Five other markets. One shot each.

A rule that only works on one asset is probably a quirk of that asset's history. One that works on several is more likely to be a real market behaviour.

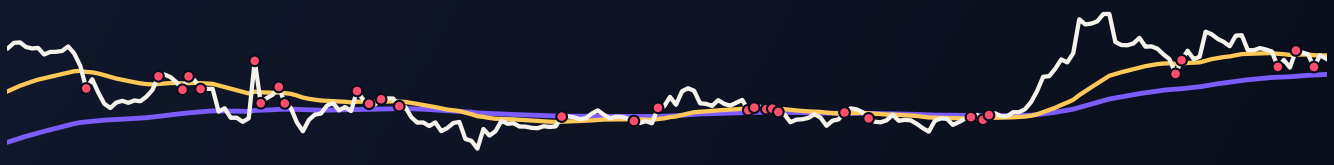
IN PLAIN ENGLISH · CROSS-SECTIONAL VALIDATION

Same rule, same settings, different markets. No re-tuning allowed - the moment you adjust settings per asset you are back to data mining with extra steps.

ETH	+42.6% CAGR	-58.3% MAX DD	$179,334 FROM $10k	$36,522 HOLDING	PASS
BNB	+75.8% CAGR	-56.8% MAX DD	$867,156 FROM $10k	$515,367 HOLDING	PASS
DOGE	+95.2% CAGR	-79.1% MAX DD	$654,056 FROM $10k	$284,368 HOLDING	PASS
SOL	+50.2% CAGR	-75.7% MAX DD	$81,104 FROM $10k	$17,461 HOLDING	PASS
XRP	+12.2% CAGR	-85.5% MAX DD	$23,487 FROM $10k	$34,148 HOLDING	FAIL

Why XRP broke it

XRP chopped sideways rather than trending. Price crossed back and forth through the 34-day average without committing to a direction. Every crossing is a trade, and every trade in chop is a small loss.



XRP daily closes. Each red dot is a crossing of the 34-day average - 31 in this stretch alone.

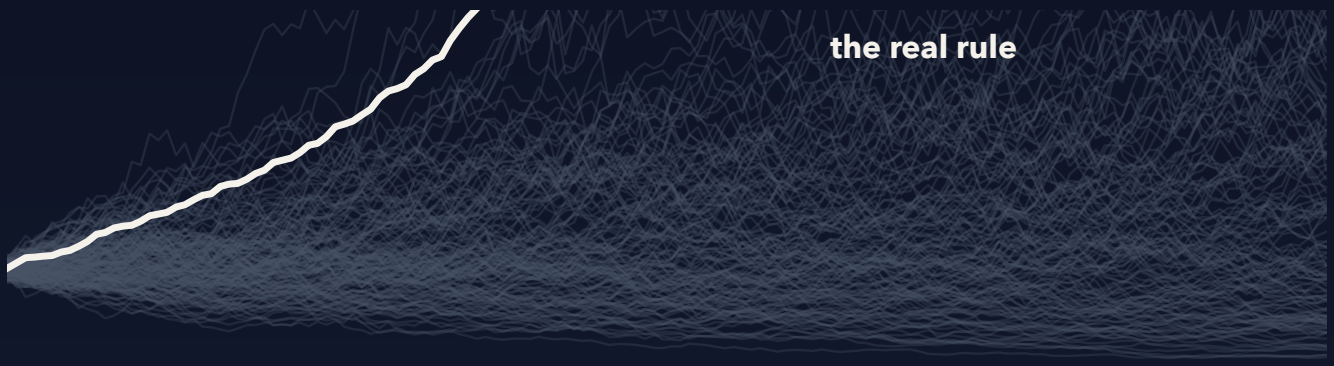
STAGE 04 · TEST 4 OF 5

500 copies with random timing.

Bitcoin rose a great deal over the test period. Any rule that holds it for a while will make money. This asks whether the TIMING added anything.

IN PLAIN ENGLISH · RANDOM-TIMING BENCHMARK

Take the real rule's number of trades and its total time in the market, then scatter those holding periods at random dates. Do it 500 times. If random dates do as well as your signal, your signal was never the edge - the market's drift was.



Illustrative shape: the grey band is 160 random-timing runs, the white line is the rule.

$16,488
RANDOM, MEDIAN
the typical random run

$211,445
RANDOM, BEST OF 500
the luckiest random run

$338,534
THE REAL RULE
beat all 500 of them

p < 0.0001

Not one of the 500 random-timing copies beat the rule. The timing is doing real work.

This is the test an earlier version failed.

An earlier long-and-short version was checked the same way and 31.5% of random runs beat it. That failure is what removed the short side.

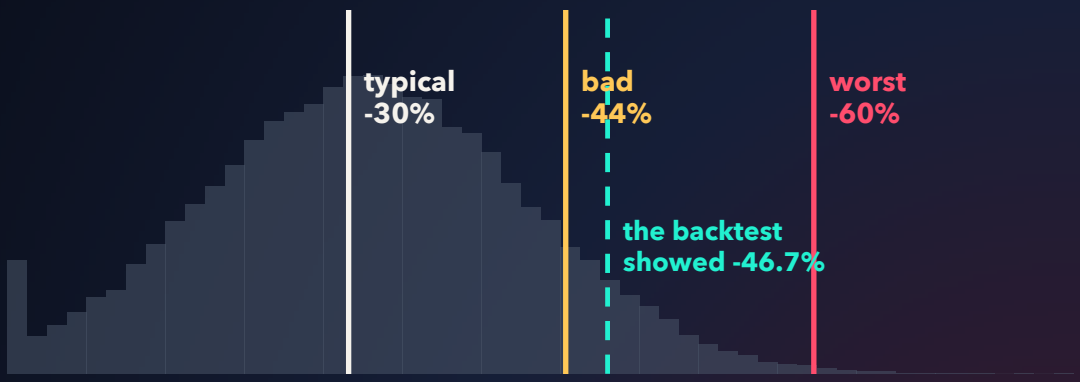
STAGE 04 · TEST 5 OF 5

5,000 reshuffles of the same trades.

A backtest gives you one ordering of your trades. You will get a different one. This asks how bad the ordering could reasonably have been.

IN PLAIN ENGLISH · MONTE CARLO

Keep every trade exactly as it was, but deal them in a different order, 5,000 times. The total return barely changes. The drawdown changes enormously - because a drawdown is just a run of losses landing together, and order decides that.



Illustrative distribution. The three marked values are the measured percentiles.

-30% TYPICAL
what a median ordering gives you

-44% BAD
a 1-in-20 ordering. Plan for this one

-60% WORST
the worst of 5,000 orderings

Why this matters more than the return

The -46.7% was one particular sequence of the same 83 trades. Half of all orderings are worse than -30%, and one in twenty worse than -44%. Nobody is stopped out by a CAGR; they are stopped out by a drawdown lasting longer than their patience.

STAGE 05

Every number.

The complete validation run. The selection row is in-sample and flatters itself; judge the rule on the six rows below it.

MARKET	TRADES	WIN	CAGR	MAX DD	SHARPE	GREW TO	HOLDING
BTC '18-26 selection set	83	30.1%	+54.2%	-46.7%	1.31	$338,534	$94,079
BTC '14-17 never searched	36	33.3%	+72.9%	-39.6%	1.46	$71,535	$58,596
ETH	95	25.3%	+42.6%	-58.3%	0.97	$179,334	$36,522
BNB	93	28.0%	+75.8%	-56.8%	1.26	$867,156	$515,367
DOGE	58	36.2%	+95.2%	-79.1%	1.01	$654,056	$284,368
SOL	62	27.4%	+50.2%	-75.7%	0.99	$81,104	$17,461
XRP lost to buy & hold	116	19.0%	+12.2%	-85.5%	0.49	$23,487	$34,148

Six of the seven beat buy-and-hold. XRP did not.

1.31 to 1.00

SHARPE, IN THEN OUT

Performance degraded from selection to validation, which is normal and expected. 1.00 is the number to trust.

+74.4%

EQUAL-WEIGHT PORTFOLIO

The rule run on BTC, ETH, BNB, DOGE and SOL together: Sharpe 1.52, max drawdown -50.3%.

~10 a year

TRADE FREQUENCY

About 43% of days in the market. Most of the work is waiting.

STAGE 05

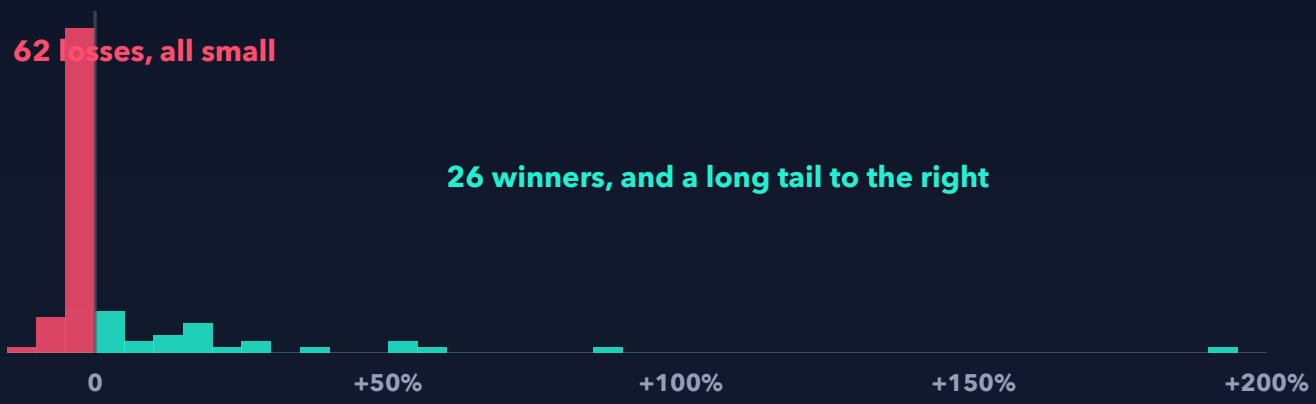
Every trade.

All 88 of them, reconstructed from the rule itself. Three in ten made money. That is not the flaw - it is the mechanism.

IN PLAIN ENGLISH · HOW A 30% WIN RATE MAKES MONEY

Losers are cut in about four days for roughly 3%. Winners are left alone for about six weeks and average 27%. You lose small often and win large rarely, and the arithmetic works out.

26 WINNERS	**62** LOSERS	**29.5%** WIN RATE
+26.7% AVERAGE WINNER	**-2.9%** AVERAGE LOSER	**9.2x** SIZE RATIO
41 days WINNER HELD	**4 days** LOSER HELD	**+192.7%** BEST TRADE



Every trade's return, bucketed in 5% steps. The losses pile up in one narrow column just left of zero; the gains stretch across the page. Removing the five best trades removes most of the profit - which is exactly why there is no take-profit rule.

Reconstructed from 30-minute BTCUSDT bars re-cut into IST days, then run through the rule exactly as written. It produces 88 trades and $327,871 against the published 83 and $338,534 - a 3% difference from data vintage and one still-open position. The published figures are kept as headline throughout this report.

STAGE 05 · TRADES 1-30 OF 88

The full list.

Entry and exit are the next day's open, exactly as the rule specifies. Equity compounds from $10,000.

#	IN	OUT	DAYS	ENTRY	EXIT	RETURN	EQUITY
1	2018-04-25	2018-04-26	1	$9,371	$9,018	-3.84%	$9,616
2	2018-04-29	2018-05-02	3	$9,370	$8,990	-4.13%	$9,219
3	2018-05-04	2018-05-09	5	$9,670	$9,199	-4.95%	$8,763
4	2018-05-10	2018-05-11	1	$9,340	$9,117	-2.46%	$8,547
5	2018-07-19	2018-07-20	1	$7,489	$7,430	-0.86%	$8,473
6	2018-07-23	2018-08-02	10	$7,518	$7,556	+0.42%	$8,509
7	2018-09-02	2018-09-06	4	$7,196	$6,903	-4.15%	$8,156
8	2018-10-16	2018-10-17	1	$6,844	$6,730	-1.74%	$8,015
9	2019-03-28	2019-06-07	71	$4,023	$7,484	+85.96%	$14,904
10	2019-06-08	2019-07-15	37	$8,069	$10,471	+29.71%	$19,332
11	2019-07-16	2019-07-17	1	$10,743	$9,688	-9.89%	$17,419
12	2019-07-19	2019-07-20	1	$10,532	$10,383	-1.50%	$17,159
13	2019-07-21	2019-07-22	1	$10,886	$10,504	-3.59%	$16,543
14	2019-08-03	2019-08-15	12	$10,459	$10,203	-2.53%	$16,125
15	2019-08-20	2019-08-22	2	$10,698	$10,117	-5.51%	$15,237
16	2019-09-04	2019-09-10	6	$10,635	$10,260	-3.61%	$14,688
17	2019-09-13	2019-09-14	1	$10,329	$10,226	-1.07%	$14,530
18	2019-09-15	2019-09-16	1	$10,367	$10,297	-0.75%	$14,421
19	2019-10-27	2019-11-09	13	$9,108	$8,813	-3.31%	$13,944
20	2020-01-08	2020-01-10	2	$8,040	$7,813	-2.90%	$13,541
21	2020-01-11	2020-02-26	46	$8,090	$9,361	+15.64%	$15,659
22	2020-04-27	2020-06-20	54	$7,614	$9,300	+22.07%	$19,114
23	2020-06-22	2020-06-25	3	$9,335	$9,320	-0.23%	$19,070
24	2020-07-07	2020-07-10	3	$9,299	$9,217	-0.96%	$18,888
25	2020-07-14	2020-07-15	1	$9,289	$9,253	-0.46%	$18,801
26	2020-07-22	2020-08-28	37	$9,376	$11,206	+19.45%	$22,457
27	2020-08-29	2020-09-03	5	$11,501	$11,354	-1.35%	$22,154
28	2020-09-17	2020-09-19	2	$11,069	$10,853	-2.03%	$21,705
29	2020-09-20	2020-09-22	2	$11,041	$10,407	-5.82%	$20,441
30	2020-09-29	2020-09-30	1	$10,873	$10,735	-1.34%	$20,168

STAGE 05 · TRADES 31-60 OF 88

The full list, continued.

#	IN	OUT	DAYS	ENTRY	EXIT	RETURN	EQUITY
31	2020-10-09	2021-01-25	108	$10,860	$31,792	+192.67%	$59,024
32	2021-01-26	2021-01-28	2	$33,958	$30,088	-11.47%	$52,254
33	2021-01-29	2021-03-01	31	$32,194	$43,582	+35.30%	$70,698
34	2021-03-02	2021-03-26	24	$48,434	$51,729	+6.73%	$75,454
35	2021-03-27	2021-04-19	23	$53,686	$55,030	+2.43%	$77,287
36	2021-05-01	2021-05-05	4	$56,652	$54,140	-4.51%	$73,801
37	2021-05-06	2021-05-13	7	$57,319	$54,369	-5.22%	$69,948
38	2021-07-27	2021-07-28	1	$39,678	$37,729	-4.99%	$66,459
39	2021-07-29	2021-08-04	6	$39,539	$37,915	-4.18%	$63,680
40	2021-08-05	2021-09-09	35	$39,624	$46,304	+16.78%	$74,368
41	2021-09-10	2021-09-11	1	$46,750	$45,791	-2.13%	$72,787
42	2021-09-15	2021-09-21	6	$46,436	$43,586	-6.21%	$68,265
43	2021-10-02	2021-11-17	46	$47,681	$60,372	+26.54%	$86,383
44	2022-03-25	2022-04-07	13	$43,886	$43,429	-1.12%	$85,418
45	2022-04-08	2022-04-09	1	$43,671	$42,693	-2.31%	$83,443
46	2022-11-06	2022-11-08	2	$21,292	$20,727	-2.73%	$81,167
47	2023-01-13	2023-02-14	32	$18,648	$21,585	+15.68%	$93,891
48	2023-02-15	2023-03-04	17	$22,112	$22,382	+1.15%	$94,971
49	2023-03-14	2023-04-22	39	$24,147	$27,936	+15.62%	$109,802
50	2023-04-27	2023-05-02	5	$29,789	$28,213	-5.37%	$103,910
51	2023-05-03	2023-05-09	6	$28,596	$27,520	-3.84%	$99,921
52	2023-05-29	2023-06-01	3	$27,597	$27,074	-1.97%	$97,950
53	2023-06-20	2023-07-21	31	$26,884	$29,600	+10.03%	$107,773
54	2023-07-22	2023-07-25	3	$29,997	$29,066	-3.18%	$104,346
55	2023-08-09	2023-08-11	2	$29,881	$29,444	-1.54%	$102,744
56	2023-10-03	2023-10-12	9	$28,005	$26,638	-4.96%	$97,650
57	2023-10-17	2024-01-14	89	$28,367	$42,845	+50.96%	$147,417
58	2024-01-16	2024-01-17	1	$43,155	$42,966	-0.51%	$146,661
59	2024-01-30	2024-04-14	75	$43,005	$66,875	+55.43%	$227,957
60	2024-04-23	2024-04-25	2	$66,474	$64,694	-2.75%	$221,681

STAGE 05 · TRADES 61-88 OF 88

The full list, continued.

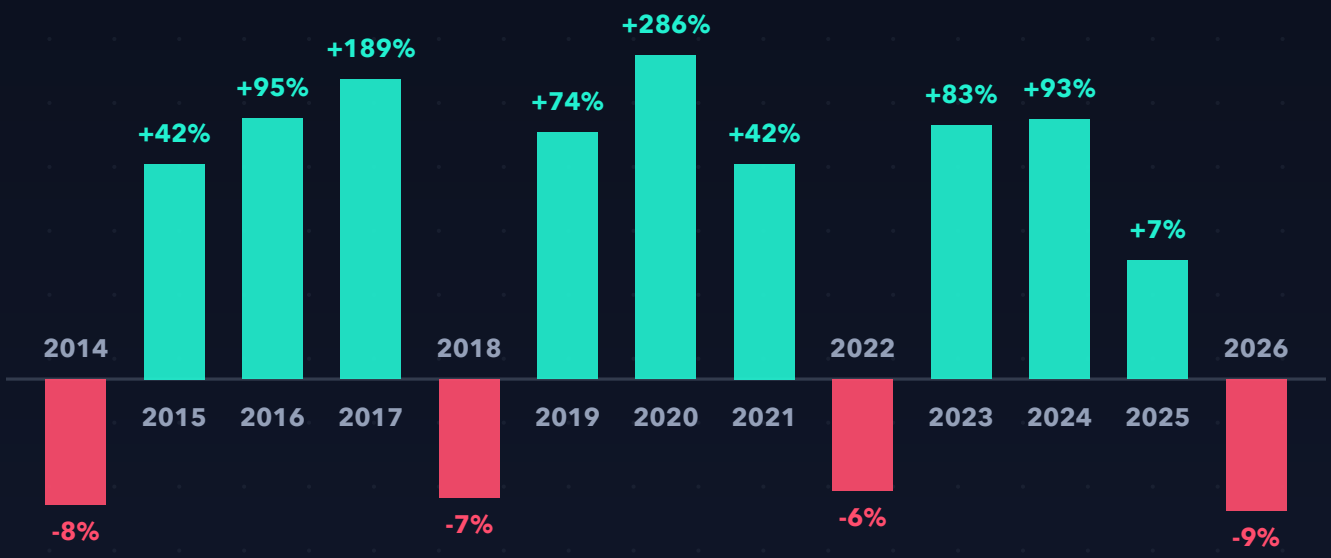
#	IN	OUT	DAYS	ENTRY	EXIT	RETURN	EQUITY
61	2024-05-16	2024-06-12	27	$65,350	$66,995	+2.44%	$227,094
62	2024-06-13	2024-06-14	1	$69,196	$66,868	-3.44%	$219,284
63	2024-07-16	2024-08-02	17	$63,442	$63,070	-0.66%	$217,833
64	2024-08-24	2024-08-28	4	$63,125	$61,893	-2.03%	$213,419
65	2024-09-18	2024-09-19	1	$61,002	$60,520	-0.86%	$211,574
66	2024-09-20	2024-10-03	13	$63,604	$61,178	-3.89%	$203,344
67	2024-10-05	2024-10-10	5	$62,150	$61,434	-1.23%	$200,849
68	2024-10-12	2024-12-23	72	$62,742	$96,185	+53.23%	$307,758
69	2024-12-25	2024-12-27	2	$98,457	$95,916	-2.66%	$299,582
70	2025-01-03	2025-01-09	6	$96,903	$94,921	-2.12%	$293,231
71	2025-01-15	2025-01-28	13	$96,642	$99,171	+2.54%	$300,685
72	2025-01-29	2025-02-03	5	$102,855	$97,832	-4.96%	$285,776
73	2025-02-21	2025-02-22	1	$98,281	$97,075	-1.30%	$282,057
74	2025-04-23	2025-06-18	56	$91,200	$103,681	+13.61%	$320,446
75	2025-06-25	2025-08-02	38	$105,845	$114,155	+7.78%	$345,364
76	2025-08-05	2025-08-06	1	$115,274	$113,440	-1.67%	$339,610
77	2025-08-07	2025-08-19	12	$115,427	$116,361	+0.73%	$342,105
78	2025-08-23	2025-08-24	1	$116,885	$115,073	-1.63%	$336,545
79	2025-09-11	2025-09-23	12	$113,390	$112,675	-0.71%	$334,170
80	2025-09-30	2025-10-12	12	$113,963	$111,894	-1.89%	$327,853
81	2025-10-27	2025-10-30	3	$113,497	$111,433	-1.89%	$321,645
82	2026-01-15	2026-01-17	2	$97,041	$95,209	-1.96%	$315,333
83	2026-04-18	2026-04-20	2	$77,198	$74,811	-3.17%	$305,345
84	2026-04-21	2026-04-30	9	$76,102	$75,150	-1.33%	$301,298
85	2026-05-01	2026-05-19	18	$76,281	$76,355	+0.02%	$301,362
86	2026-05-22	2026-05-23	1	$77,742	$76,807	-1.28%	$297,512
87	2026-05-26	2026-05-27	1	$77,638	$76,101	-2.06%	$291,398
88	2026-08-20	2026-09-02 *	13	$68,280	$76,877	+12.52%	$327,871

*** Trade 88 was still open at the close on 2 September 2026. Its return is marked to that close, not realised.**

STAGE 05

Thirteen years, year by year.

Ten of thirteen positive. The worst calendar year cost 9%.



Bars are on a compressed scale so 2020's +286% does not flatten the rest.

The losing years are the filter working

2018 cost 7%, 2022 cost 6%, and 2026 has cost 9% so far. In each of those years Bitcoin itself fell far more - 2022 alone was -64%. A trend rule that sits flat through a crash does not make money in it; it simply avoids most of the damage. That is the trade being made, and it is also why the rule loses to buy-and-hold in a strong bull year.

What was tested and thrown away

- x **The short side**
  +2.4% across nine years while adding 23 points of drawdown. Removed.
- x **Fixed stop losses**
  0.5%, 1% and 2% all made results worse - they cut winners at noise level.
- x **Take-profit targets**
  The whole return comes from a few positions held past 30 days. A target removes exactly those.
- x **100/21 instead of 100/34**
  Looked best on the selection data, then failed to replicate out-of-sample. Chasing the peak is chasing noise.

THE OUTLIER

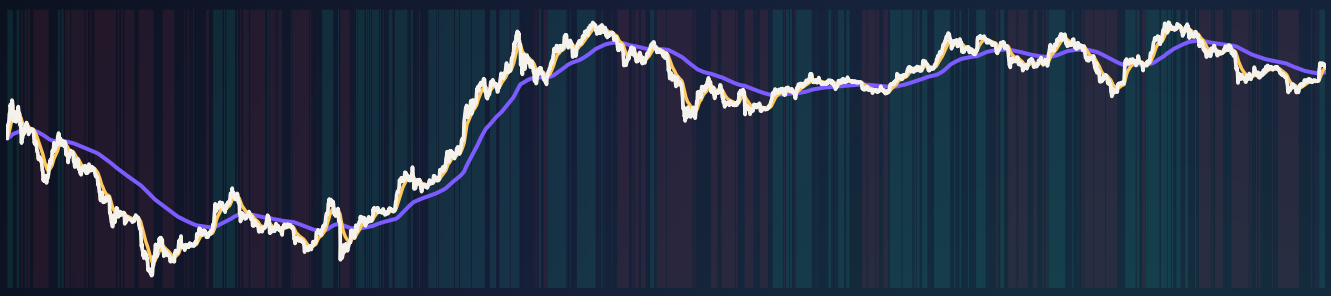
The highest return of everything tested.

Same idea, different market, and the short side switched back on. This is the single best result any configuration produced anywhere in the search.

ETH · 200 EMA / 21 EMA · long + short

$578,181	+64.7%	-44.5%
FROM $10,000	CAGR	MAX DD
1.18	**187**	**32.1%**
SHARPE	TRADES	WIN RATE

8.1 years, June 2018 to July 2026. Buy-and-hold over the same window turned $10,000 into $36,522 - the rule returned roughly sixteen times that.



ETH daily closes, log scale. Purple = 200 EMA, amber = 21 EMA. Teal = long, red = short, dark = flat.

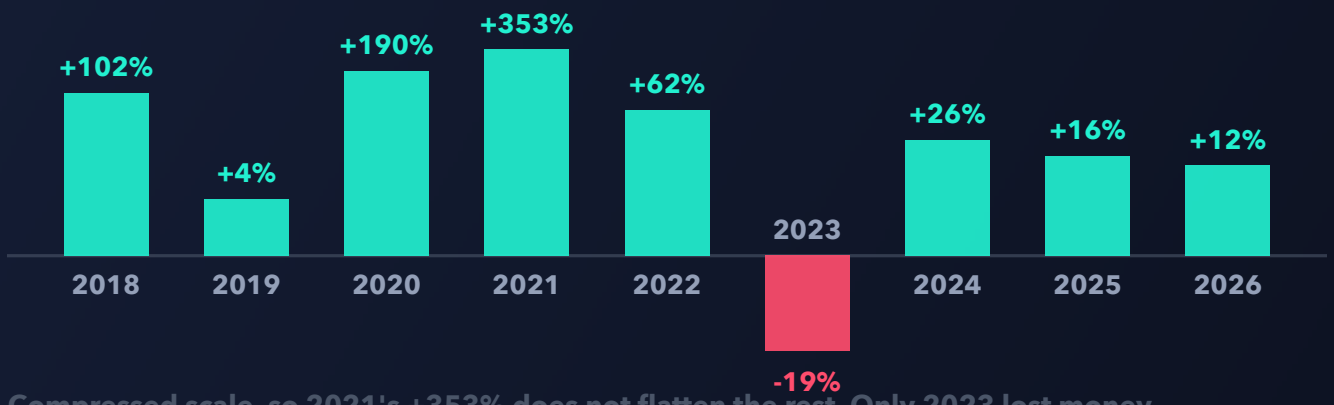
Where the rules differ from the Bitcoin version

MARKET	ETH/USDT
INDICATORS	EMA(200) filter and EMA(21) exit - slower filter, faster exit
ENTER SHORT	close < EMA(200) AND close < EMA(21)
EXIT SHORT	close > EMA(21) OR close > EMA(200)
UNCHANGED	Daily IST close, next open, full balance, 1x, no stop, no target

Rules 7 and 9 are the flatten-on-regime-flip version: either average closes the trade, not just the fast one.

What that return actually costs.

Eight of nine years positive, and a drawdown profile far worse than the headline suggests.



Compressed scale, so 2021's +353% does not flatten the rest. Only 2023 lost money.

On ETH, the short side actually pays

LONGS	103 trades	31% win	+4.7% average
SHORTS	84 trades	33% win	+1.9% average

That is the opposite of Bitcoin, where shorts returned +2.4% across nine years and added 23 points of drawdown. Average winner +19.0% held for 26 days; average loser -3.9% held for 4. In the market 67% of the time.

Two numbers to size against

$129,841 EXCLUDING 2021

Strip out the best year entirely and $10,000 still becomes $129,841 - a 13x. This is not one lucky year the way DOGE was.

-51.6% / -68.7% MEDIAN / BAD DRAWDOWN

Across 5,000 reshuffles. Worst was -84.3%. The -44.5% in the backtest is the optimistic end - plan for losing half the account.

HIGHEST RETURN, NOT BEST TESTED

It never went through the five tests in this report, and it is long-and-short - the direction filter that failed on Bitcoin. Best number found, not the safest.

STAGE 06

What this still does not prove.

Five tests passed is not the same as the rule working. Here is everything that is still wrong with it.

1 The sample is small

83 trades across 8 years, and because trades cluster inside the same trends, that is closer to a dozen genuinely independent events. Statistics on a dozen events are weak however the arithmetic is presented.

2 XRP broke it outright

Not underperformance - a loss, while doing nothing made money. Any market that chops sideways through the averages will do the same thing.

3 There has never been a lost decade

Crypto's entire history is one long uptrend with violent interruptions. A trend-following rule has never faced a market that simply goes nowhere for ten years. That is the untested failure mode, and it is the one that would hurt most.

4 The validation data is spent

It was opened once, as it should be. Every further idea tested against it from now on is in-sample, and its results mean less.

5 Drawdowns are survivable on paper only

-44% is easy to accept in a table and very hard to sit through for months with real money. Most systems fail at the person, not the arithmetic.

Supported. Not proven.

Different words, and the difference is the whole point of this document.